

QUESTION 1

What is the sum of $3E6_{16}$ and 444_{16} ?

- A. 720_{16} B. 777_{16} C. $7IA_{16}$ D. $83A_{16}$ E. $82A_{16}$

QUESTION 2

Which of these declares a variable to have the type `MyClass` and initializes it using the two parameter constructor with parameters 3 and 9?

- A. `(MyClass)m = MyClass(3,9)`
 B. `new MyClass m(3,9)`
 C. `MyClass m = 3, 9`
 D. `MyClass m = new MyClass(3,9)`
 E. `MyClass m = MyClass(3,9)`

```
public class MyClass {
    public MyClass(int x, int y) {
        xdata = x; ydata = y;
    }

    public MyClass(int x) {
        xdata = x;
    }
}
```

QUESTION 3

If a `MyClass` object is built using the second constructor, to what is its data member `ydata` initialized?

- A. Same as `xdata` B. 65535
 C. 0 D. Not initialized
 E. Syntax error to not initialize

```
// methods not shown

private int xdata, ydata;
}
```

QUESTION 4

How many '*'s are output by the code to the right?

- A. 160 B. 40 C. 26
 D. 10 E. 0

```
for (int i=0; i<10; ++i)
    for (int j=1; j<16; j*=2)
        System.out.print("*");
```

QUESTION 5

What replaces `<*1>` in the code to the right to give the character at position `i` in `s`, converted to lower case?

- A. `if (s.charAt(i) >= 'A' && s.charAt(i) <= 'Z') s.charAt(i) - 'A'`
 B. `Character.isLowerCase?s[i]:s[i] - 'a'`
 C. `Character.toLowerCase(s[i])`
 D. `Character.toLowerCase(s.charAt(i))`
 E. More than one of these

```
public static void smallOutput(String s) {
    for (int i=0; i<s.length(); ++i)
        System.out.print(<*1>);
}
```

QUESTION 6

What does `int[][] a` look like after the static method call `process(a)` when `a` begins as the matrix below?

1	4	7
2	5	8
3	6	9

A.

1	2	3
4	5	6
7	8	9

B.

1	5	9
3	7	11
5	9	13

C.

1	4	7
3	7	11
7	12	13

D.

1	3	5
4	7	9
7	12	13

E.

0	1	2
1	2	3
2	3	4

```
public static void process(int[][] a) {
    for (int i=0; i<a.length; ++i)
        for (int j=0; j<a[i].length; ++j)
            a[i][j] = a[j][i] + i + j;
}
```

QUESTION 7

Which of these can replace `<*1>` in the code to the right to give a valid initialization to `b`?

- A. 17
- B. x
- C. (boolean)x
- D. boolean(x)
- E. None of these

```
int x = 17;
boolean b = <*1>;
```

QUESTION 8

Which of these statements sets `b` to `true`?

- A. `b = (x < y) && !(z < y);`
- B. `b = !(x < y) || !(y < z);`
- C. `b = !( (x > y) || (y < z) );`
- D. `b = !( !(x > y) && (y < z) );`
- E. `b = x < y < z;`

```
int x = 1, y = 2, z = 3;
boolean b;
```

QUESTION 9 What is returned by f("UIL")? A. 38 B. 39 C. 40 D. 41 E. 42	<pre>public static int f(String s) { int x = 0, y = 0; while (x < s.length()) { y += s.charAt(x) - 'A'; ++x; } return y; }</pre>
QUESTION 10 Which of these data structures provides insertion access only at one end of a list and removal access only at the other end? A. queue B. stack C. heap D. tree E. map	
QUESTION 11 What replaces <*1> so that the Scanner will use all non-digits to delimit tokens? A. "0-9" B. "^0-9" C. "[^0-9]" D. "^0-^9" E. "-[0-9]"	<pre>Scanner s = new Scanner(System.in); s.useDelimiter(<*1>);</pre>
QUESTION 12 What is output by the code to the right? A. 22 B. 31 C. 23 D. 4 E. Nothing	<pre>int[] a = {1, 2, 3, 4, 5}; int i = 0; int j = i++; System.out.print(""+a[++i]+a[j++]);</pre>
QUESTION 13 What replaces <*1> in the code to the right to add the number 27 to a? A. 27 B. (Integer)27 C. new Integer(27) D. Either A or C E. A, B, or C	<pre>ArrayList<Integer> a = new ArrayList<Integer>(); a.add(<*1>);</pre>
QUESTION 14 What is output by the code to the right? A. hello B. goodbye C. false D. true E. Nothing	<pre>Set<String> s = new TreeSet<String>(); s.add("hello"); s.add("hello"); s.add("goodbye"); s.remove("hello"); System.out.print(s.contains("hello"));</pre>

QUESTION 15

What replaces **<*1>** in the code to the right to call the other static method `mystery()` with the parameters `a`, `rank`, the index of the first element of `a`, and one more than the index of the last element of `a`?

- A. `mystery(a, rank, 1, length)`
- B. `mystery(a, rank, 1, 100)`
- C. `mystery(a, rank, 1, a.length())`
- D. `mystery(a, rank, 0, a.length())`
- E. `mystery(a, rank, 0, a.length)`

QUESTION 16

Assume **<*1>** is filled in correctly. What does `mystery()` do?

- A. Returns the value corresponding to position `rank` in `a` if the elements of `a` were sorted without sorting all of `a`
- B. Sorts `a` and returns the value corresponding to position `rank`
- C. Looks for a position in `a` corresponding to value `rank`
- D. Returns the median element of `a`
- E. Returns the sum of the elements of `a` except for the element at position `rank`

QUESTION 17

What is the expected running time of `mystery(a, 0)` when `a` is a random array with `n` elements?

- A. $O(1)$
- B. $O(\log n)$
- C. $O(n)$
- D. $O(n^2)$
- E. $O(n \log n)$

QUESTION 18

What is output by the code to the right?

- A. `scienc ecomputer`
- B. `scienceco mputer`
- C. `computer science`
- D. `computers cience`
- E. Nothing

```
public static void swap(int[] a, int i,
                        int j) {
    int temp = a[i];
    a[i] = a[j];
    a[j] = temp;
}

public static int partition(int[] a,
                           int front, int back) {
    int pivot = a[front];
    int i = front+1;
    int j = back-1;
    while (i<j) {
        while((i<back) && (a[i] < pivot)) ++i;
        while((j>front) && (a[j] >= pivot)) --j;

        if (i<j) swap(a,i,j);
    }
    swap(a,front,j);
    return j;
}

public static int mystery(int[] a,
                          int rank) {
    return <*1>;
}

public static int mystery(int[] a,
                          int rank, int front, int back) {
    if ((front > rank) || (back <= rank))
        throw new IllegalArgumentException();
    int mid = partition(a,front,back);
    if (mid == rank)
        return a[mid];
    else if (mid > rank)
        return mystery(a,rank,front,mid);
    else
        return mystery(a,rank,mid+1,back);
}
```

```
String s1 = "computer", s2 = "science";
StringBuffer sb = new StringBuffer(s1);
sb.append(s2);
sb.insert(9, " ");
System.out.print(sb);
```

QUESTION 19

What replaces **<*1>** in the code to the right to indicate that `numPairs` is a class variable, with one value shared by all instances of `Shoe`?

- A. `void` B. `static`
- C. `unique` D. `shared`
- E. `catch`

For the remaining questions, assume that **<*1>** has been filled in correctly.

QUESTION 20

What is output by the code below?

```
Shoe bigShoe = new Shoe("loafer",
                        "brown", 15, "EEE");
System.out.print(bigShoe);
```

- A. `brown loafer size: 15EEE`
- B. `loafer brown size:15EEE`
- C. `brownloafer size:15EEE`
- D. The memory address of `bigShoe`
- E. An exception is thrown

QUESTION 21

What is output by the code below?

```
Shoe otherShoe = new Shoe();
System.out.print(otherShoe);
```

- A. `size:`
- B. `color type size: sizewidth`
- C. At compile time the call to `Shoe()` will be an error
- D. At run time the call to `Shoe()` will throw an exception
- E. At run time the call to `print()` will throw an exception

QUESTION 22

What is the superclass of `Shoe`?

- A. `String` B. `int`
- C. `Object` D. `Clothing`
- E. It has no superclass

```
public class Shoe {
    public Shoe(String type, String color,
                int size, String width) {
        this.type = type;
        this.color = color;
        this.size = size;
        this.width = width;
        ++numPairs;
    }

    public String toString() {
        return color + " " + type + " size: "
            + size + width;
    }

    private String type, color;
    private int size;
    private String width;
    private <*1> int numPairs = 0;
}
```

QUESTION 23

Which of these declares and creates an object that associates Book objects with Person objects?

- A. `List<Book, Person> obj = new ArrayList<Book, Person>();`
- B. `Map<Book, Person> obj = new TreeMap<Book, Person>();`
- C. `List<Book, Person> obj = new List<Book, Person>();`
- D. `Map<Book, Person> obj = new Map<Book, Person>();`
- E. Either A or C

```
public class Person {
    // constructors and methods not shown
    private String familyName, givenName;
}

public class Book {
    // constructors and methods not shown
    private String title, genre;
    private int numPages;
}
```

QUESTION 24

Assume that the object in the previous question has been created and that items have been added to it. What type of object is returned by `obj.iterator().next()`?

- A. Book
- B. Person
- C. Map.Entry
- D. Iterator
- E. There is no `iterator()` method for `obj`

QUESTION 25

Which of these best describes the value `z` after executing the code to the right? (Assume the input actually contains a floating point value.)

- A. The largest possible long value
- B. The smallest possible long value
- C. The largest integer which is not more than the floating point value read from the input
- D. The smallest integer which is at least as big as the floating point value read from the input
- E. The integer closest to the floating point value read from the input

```
Scanner s = new Scanner(System.in);
long z = Math.ceil(s.nextDouble());
```

QUESTION 26

What is output by the code to the right?

- A. 123
- B. 456
- C. a
- D. abc
- E. Nothing

```
String[] strings =
    "abc123def456ghi".split("[a-z]");
System.out.print(strings[1]);
```

QUESTION 27

What is output by the code to the right?

- A. 96 B. 37 C. 36
D. 33 E. 30

```
int x = 37, y = 59;
System.out.print(x^y);
```

QUESTION 28

What does `a` look like at the point in the execution of `sort(a)` when `i` is incremented in the `for` loop from 1 to 2 when `a` begins as the array below?

14	-2	-5	81	-8	19
----	----	----	----	----	----

- A.

14	-2	-5	81	-8	19
----	----	----	----	----	----

B.

-8	-2	-5	81	14	19
----	----	----	----	----	----

C.

-8	-5	-2	81	14	19
----	----	----	----	----	----

D.

81	19	-5	14	-8	-2
----	----	----	----	----	----

E.

81	19	14	-2	-5	-8
----	----	----	----	----	----

```
public static void sort(int[] a) {
    for (int i=0; i<a.length-1; ++i) {
        int item = a[i];
        int index = i;
        for (int j=i; j<a.length; ++j)
            if (a[j] > item) {
                item = a[j];
                index = j;
            }
        int temp = a[i];
        a[i] = a[index];
        a[index]=temp;
    }
}
```

QUESTION 29

What sorting algorithm is implemented by `sort()`?

- A. Selection sort B. Insertion sort
C. Quicksort D. Merge Sort
E. Random sort

QUESTION 30

Which of the following correctly declares and initializes an iterator for `list` that can be used to both to add and remove items from the list?

- A. `ListIterator<String> iter = list.iterator();`
B. `ListIterator<String> iter = list.listIterator();`
C. `Iterator<String> iter = list.listIterator();`
D. Either A or B
E. A, B, or C

```
LinkedList<String> list =
    new LinkedList<String>();
// code to initialize list not shown
```

QUESTION 31

The heap data structure uses an `ArrayList` to hold the elements of a complete binary tree. The root is in position 0, its children are in positions 1 and 2, the next level is in positions 3, 4, 5, and 6, and so on. What replaces `<*1>`, `<*2>`, and `<*3>` in the code to the right to give the index of a node's left child, right child, and parent, respectively?

- A. `<*1>`: 0
`<*2>`: 1
`<*3>`: 2
- B. `<*1>`: $2 * \text{index}$
`<*2>`: $2 * (\text{index} + 1)$
`<*3>`: $\text{index} / 2$
- C. `<*1>`: $2 * \text{index} + 1$
`<*2>`: $2 * \text{index} + 2$
`<*3>`: $(\text{index} - 1) / 2$
- D. `<*1>`: $2 * \text{index}$
`<*2>`: $2 * \text{index} + 1$
`<*3>`: $\text{index} / 2$
- E. `<*1>`: $2 * \text{index} - 1$
`<*2>`: $2 * \text{index}$
`<*3>`: $(\text{index} + 1) / 2$

QUESTION 32

What property is maintained by the `Heap` class?

- A. The smallest value is at the root
- B. The largest value is at the root
- C. All subtrees have the heap property
- D. Both A and C
- E. Both B and C

QUESTION 33

What is the running time of removing an item from a heap containing n elements?

- A. $O(1)$ B. $O(\log n)$ C. $O(n)$
- D. $O(n^2)$ E. $O(n \log n)$

QUESTION 34

Which of these classes could be used as elements of a `Heap` object?

- A. `String` B. `Double`
- C. `Random` D. Both A and B

```
public class Heap <E extends Comparable> {
    public void add(E item) {
        data.add(item);
        int index = data.size()-1;
        while ( (index != 0) &&
                (data.get(parent(index)).
                 compareTo(item)<0) ) {
            data.set(index,
                    data.get(parent(index)));
            data.set(parent(index), item);
            index = parent(index);
        }
    }

    public E remove() {
        E item = data.get(0);
        data.set(0,
                data.remove(data.size()-1));
        int index = 0;
        while (left(index) < data.size()) {
            int left = left(index),
                right = right(index);
            int swapindex;
            if (right >= data.size())
                swapindex = left;
            else if (data.get(left).compareTo(
                    data.get(right)) > 0)
                swapindex = left;
            else swapindex = right;
            if (data.get(index).compareTo(
                    data.get(swapindex)) < 0) {
                E temp = data.get(index);
                data.set(index,
                        data.get(swapindex));
                data.set(swapindex, temp);
                index = swapindex;
            }
            else index = data.size();
        }
        return item;
    }

    private static int left(int index) {
        return <*1>;
    }
    private static int right(int index) {
        return <*2>;
    }

    private static int parent(int index) {
        return <*3>;
    }

    private ArrayList<E> data =
        new ArrayList<E>();
}
```

E. A, B, and C	
-------------------	--

}

QUESTION 35

What is returned by $f(0, 4)$?

- A. 4 B. 5 C. 6
D. 10 E. 100

QUESTION 36

If x is a positive integer, what is returned by $f(2, x)$?

- A. Three more than twice x
B. Two more than three times x
C. The square of x
D. Two more than the square of x
E. One less than the square of $(x+1)$

QUESTION 37

What is output by the code to the right?

- A. 3.1416 B. 3.14159265
C. 3.141 D. 3.142
E. 3.14

```
public static int f(int x, int y) {
    if (x==0) return 1+y;
    if (y==0) return f(x-1,1);
    return f(x-1, f(x,y-1));
}
```

```
double pi = 3.14159265;
System.out.printf("%2.4f", pi);
```

QUESTION 38

Which of these statements is true about `process()`?

- A. It can throw `IOException` and its subclasses
B. It can throw `RuntimeException` and its subclasses
C. It can throw `Error` and its subclasses
D. Both A and B
E. A, B, and C

```
public static int process(String s)
                        throws IOException {
    // code not shown
}
```

QUESTION 39

What is returned by $f("abacab")$?

- A. 10 B. 12
C. 14 D. 16
E. 18

```
public static int f(String s) {
    int answer = 0;
    for (int i=0; i<s.length(); ++i)
        switch((s.charAt(i)-'a')%2) {
            case 0 : ++answer; break;
            case 1 : answer *= 2; break;
        }
    return answer;
}
```

QUESTION 40

Which of these data structures uses a function that maps an object to its position in an array?

- A. queue B. binary tree C. linked list D. hash table E. stack

Computer Science Answer Key

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1. E	11. C	21. C	31. C
2. D	12. B	22. C	32. B
3. C	13. D	23. B	33. B
4. B	14. C	24. E	34. D
5. D	15. E	25. D	35. B
6. D	16. A	26. E	36. A
7. E	17. C	27. E	37. A
8. A	18. D	28. D	38. E
9. B	19. B	29. A	39. A
10. A	20. A	30. B	40. D